

# GradelQ: Explainable Automatic Grade Change Intelligence & Setpoint Optimizer

**Theme:** Smart Automation / Industrial Process Intelligence | **Honeywell QCS Challenge**

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**Problem Statement ID:** Honeywell QCS Challenge

**Proposed Solution:** Ensemble ML + SHAP XAI Advisor

## PROPOSED SOLUTION: GRADEIQ AI INTELLIGENCE LAYER

An intelligent external advisor that enhances Honeywell QCS Machine Direction (MD) Multivariable Model Predictive Control without disrupting plant hardware loops.

- **Proactive Off-Spec Risk Prediction (Ensemble ML):** Forecasts when the primary quality variable—**Basis Weight**—is at risk of deviating **>2.5% from setpoint** up to 15 minutes prior to violation during auto grade/recipe transitions.
- **Source-Tagged Setpoint Optimization:** Recommends corrective setpoint adjustments for **Stock Flow, Filler Flow, Dryer Steam Pressure, Wire Vacuum, and Machine Speed** to dramatically reduce stabilization time and eliminate broke cull.
- **Explicit Inference Provenance:** Every recommendation on the operator display is explicitly tagged with its root inference source (e.g., *[Historical Mining: Transition A32->B18]*, *[Recipe Process Limits]*, *[AI New Correlation Discovery]*).
- **Explainable AI (XAI) Guidance:** Integrates **SHAP (SHapley Additive exPlanations)** to bridge operator skill shortages, showing newer operators exactly *\*why\** an action is recommended and which sensor parameters are causing instability.
- **Closed-Loop Operator Trust Engine:** Features a Human-in-the-Loop dashboard with dedicated **Accept/Reject controls**. All responses are audited and logged to evaluate recommendation accuracy and adaptively recalibrate the AI models.

# TECHNICAL APPROACH & MODULAR SYSTEM ARCHITECTURE

## Core Tech Stack & Methodology

- **Machine Learning:** Scikit-Learn, XGBoost, Gradient Boosting, & Random Forest voting ensemble.
- **Explainability (XAI):** SHAP TreeExplainer real-time decomposition & conditional covariance matrix mining.
- **Dashboard & Deployment:** Gradio interactive web UI, Plotly dynamics, suitable for AWS S3/RDS & EC2 Edge inference.
- **Data Engineering:** Automated trajectory feature gradients over rolling 2.5-minute transition windows.

**Data Flow Communication:** Ingests historical DCS trends, QCS scanner logs, MIS reports, and operator alarms -> Executes Kalman sliding-window feature imputation -> Predicts deviation trajectories -> Surfaces source-tagged explainable guidance.



# FEASIBILITY, INDUSTRIAL VIABILITY & EDGE CASE MITIGATION

## Why Feasible & Deployable Today

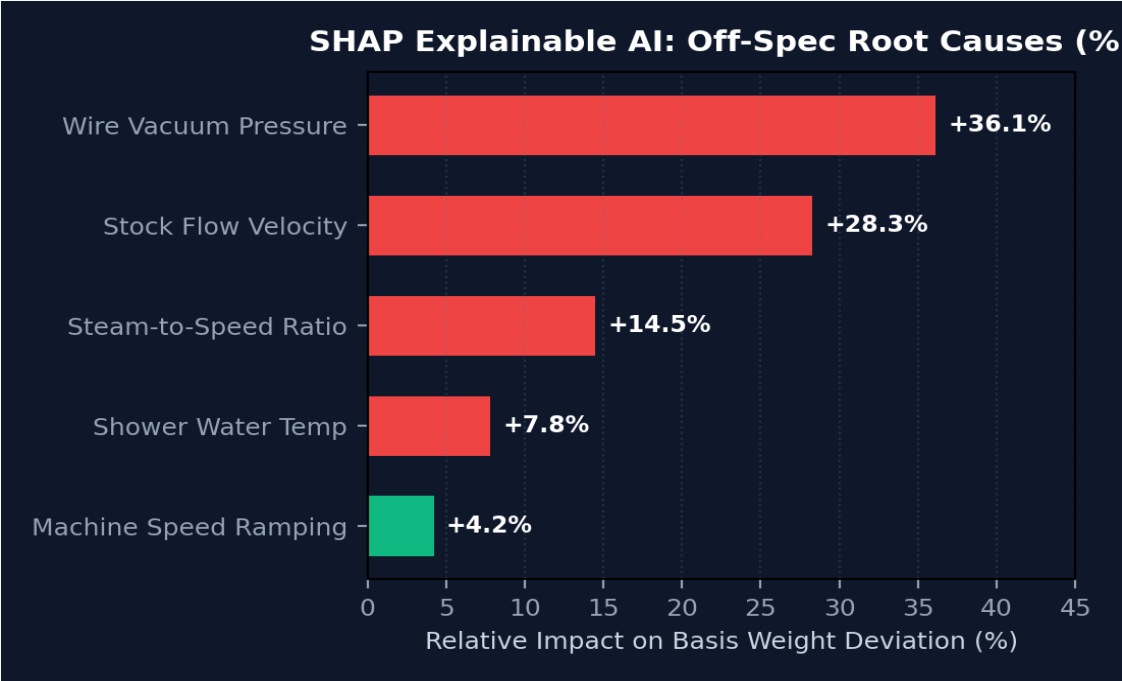
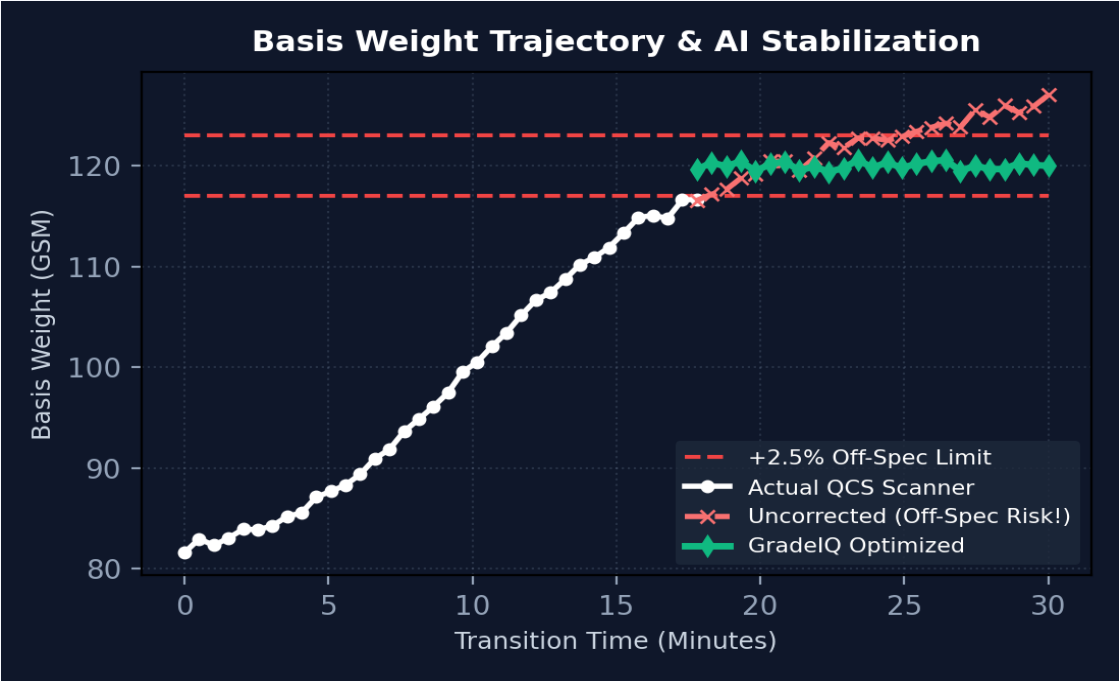
- **Zero Hardware Modification:** Sits alongside (not inside) existing Honeywell QCS MD Multivariable control loops as an external supervisory advisory layer.
- **Reuses Existing Data:** Fully leverages existing DCS historians, scanner diagnostic tables, and recipe limit logs already archived by mills.
- **High Return on Investment:** Reducing broke cull material by just 2 minutes per grade change saves mills up to \$350,000 annually in energy and wasted chemical fiber.

## Manufacturing Edge Cases & Mitigation

- **Sensor Latency & Packet Loss:** Implements real-time forward-fill linear imputation and sliding-window Kalman filtering for momentary DCS scanner drops.
- **Noisy & Outlier Scanner Data:** Robust ensemble voting algorithms prevent transient electrical anomalies from triggering false alarm deviation alerts.
- **Operator Trust & Skill Shortage:** SHAP waterfalls translate numerical logits into plain English explanations, earning junior operator confidence.
- **Sparse Rare Transitions:** Utilizes synthetic minority time-series data augmentation (SMOTE-TS) to guide extreme GSM recipe jumps.

# ARTIFACTS & REAL-TIME OPERATOR ADVISORY STUDIO

**Deliverables Repository Structure:** Includes realistic industrial dataset (*paper\_mill\_grade\_change\_data.csv*), ensemble XAI pipeline, interactive Gradio/Plotly dashboard studio, and automated Google Colab 1-click execution notebook.



## RESEARCH, REFERENCES & INDUSTRIAL BENCHMARKS

**[Ref 1] Honeywell Experion MX & QCS Automatic Grade Change Application Notes:** Product documentation on Machine Direction (MD) Multivariable Model Predictive Control, target calculation, and coordinated ramping of paper machine setpoints.

**[Ref 2] Lundberg, S. M., & Lee, S.-I. (2017):** '*A Unified Approach to Interpreting Model Predictions*' (SHAP). Advances in Neural Information Processing Systems (NeurIPS 2017). Foundation for GradelQ real-time operator root-cause explainability.

**[Ref 3] Chen, T., & Guestrin, C. (2016):** '*XGBoost: A Scalable Tree Boosting System*'. ACM SIGKDD International Conference on Knowledge Discovery and Data Mining. Used for robust multi-loop trajectory risk classification.

**[Ref 4] Ribeiro, M. T., Singh, S., & Guestrin, C. (2016):** '*Why Should I Trust You?: Explaining the Predictions of Any Classifier*' (LIME). ACM SIGKDD 2016. Benchmark reference for trust modeling in human-in-the-loop cyber-physical systems.

**[Ref 5] TAPPI Journal Case Studies on QCS Control Loop Optimization:** Published industrial engineering benchmarks demonstrating that proactive control loop decoupling and faster grade transition ramps significantly reduce broke fiber cull and thermal stabilization losses in commercial paper mills.